NeuroImage 112 (2015) 327–340

Contents lists available at ScienceDirect

NeuroImage

journal homepage: www.elsevier.com/locate/ynimg

Parametric variation of gamma frequency and power with luminance contrast: A comparative study of human MEG and monkey LFP and spike responses

A. Hadjipapas [c,d,*,1], E. Lowet [a,b,**,2], M.J. Roberts [a,b], A. Peter [e,f], P. De Weerd [a,b]

[a] Radboud University Nijmegen, Donders Institute for Brain, Cognition and Behaviour, Nijmegen, The Netherlands
[b] Maastricht University, Maastricht, The Netherlands
[c] University of Nicosia Medical School, Cyprus
[d] St George's University of London, UK
[e] Ernst Strüngmann Institute (ESI) for Neuroscience in Cooperation with Max Planck Society, Germany
[f] International Max Planck Research School for Neural Circuits, Frankfurt, Germany

ARTICLE INFO

Article history:
Accepted 27 February 2015
Available online 10 March 2015

Keywords:
Gamma oscillations
Peak frequency
Visual contrast
MEG
LFP
PING networks

ABSTRACT

Gamma oscillations contribute significantly to the manner in which neural activity is bound into functional assemblies. The mechanisms that underlie the human gamma response, however, are poorly understood. Previous computational models of gamma rely heavily on the results of invasive recordings in animals, and it is difficult to assess whether these models hold in humans. Computational models of gamma predict specific changes in gamma spectral response with increased excitatory drive. Hence, differences and commonalities between spikes, LFPs and MEG in the spectral responses to changes in excitatory drive can lead to a refinement of existing gamma models. We compared gamma spectral responses to varying contrasts in a monkey dataset acquired previously (Roberts et al., 2013) with spectral responses to similar contrast variations in a new human MEG dataset. We found parametric frequency shifts with increasing contrast in human MEG at the single-subject and the single-trial level, analogous to those observed in the monkey. Additionally, we observed parametric modulations of spectral asymmetry, consistent across spikes, LFP and MEG. However, while gamma power scaled linearly with contrast in MEG, it saturated at high contrasts in both the LFP and spiking data. Thus, while gamma frequency changes to varying contrasts were comparable across spikes, LFP and MEG, gamma power changes were not. This indicates that gamma frequency may be a more stable parameter across scales of measurements and species than gamma power. The comparative approach undertaken here represents a fruitful path towards a better understanding of gamma oscillations in the human brain.

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Introduction

In monkey visual cortex, gamma oscillations have been studied for many years, and are thought to mediate a number of computational functions (Buzsáki and Wang, 2012; Engel et al., 1999; Fries, 2009; Fries et al., 2007; Tiesinga and Sejnowski, 2010). Gamma oscillations are thought to be related to local inhibition. Because of the inhibition, excitatory neurons show low excitability in general (Buzsáki, 2006). However, the inhibition is thought to vary over time leading to a rhythmic modulation of neuronal excitability, thereby defining windows of opportunity for neurons to fire (Buzsáki and Draguhn, 2004; Csicsvari et al., 1999; Lakatos et al., 2005). The length of the information–integration phases, the spacing between them, and the duration of the firing phases, is governed by the *frequency* of the oscillation. Hence, communication among different neural populations is enabled by an appropriate and consistent relationship in time among these windows of opportunity (Fries, 2005), which requires a sufficient match in the oscillatory frequencies between communicating populations (Roberts et al., 2013). More generally, the theory of weakly coupled oscillators predicts that similar frequencies are a prerequisite for synchronization of nearby neuronal populations (Lowet et al., 2015). Thus, oscillation frequency is a network parameter of utmost functional relevance and theoretical interest (Barardi et al., 2014; Cohen, 2014; Colgin et al., 2009). This is supported by correlations between gamma peak frequency and relevant parameters including perceptual performance (Edden et al., 2009), age (Gaetz et al., 2012), and cortical structure (Schwarzkopf et al., 2012). Power is the other highly relevant feature of spectral responses. Studies in monkeys (Jia et al., 2013; Ray and Maunsell, 2010; Roberts et al.,

* Correspondence to: A. Hadjipapas, University of Nicosia Medical School, Cyprus.
** Correspondence to: E. Lowet, Radboud University Nijmegen, Donders Institute for Brain, Cognition and Behaviour, Nijmegen, The Netherlands.
E-mail addresses: hadjipapas.a@unic.ac.cy (A. Hadjipapas), e.lowet@fcdonders.ru.nl (E. Lowet).
[1] Former address: Radboud University Nijmegen, Donders Institute for Brain, Cognition and Behaviour, Nijmegen, The Netherlands
[2] Shared first authorship.

http://dx.doi.org/10.1016/j.neuroimage.2015.02.062
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2013) have shown that power tends to be maximized around a mid-level of contrast or excitatory drive, and a reduction of power at much higher or lower levels of drive is observed suggesting an optimization of resonance at those contrasts. At the local network level, such phenomena require a specific architecture of the network, and hence the study of power modulations as a function of excitatory drive provides another important probe into the underlying gamma generating mechanisms. Further, in spatially-aggregate signals power provides an indirect measure of the strength of the underlying synchronization (Hadjipapas et al., 2009; Musall et al., 2014).

Advances in MEG beamformer source reconstruction have enabled the study of gamma oscillations localized to human visual cortex (Adjamian et al., 2004; Hall et al., 2005; Hoogenboom et al., 2006). Using these methods, several MEG studies reported human gamma oscillations to exhibit marked stimulus specificity (Adjamian et al., 2004; Duncan et al., 2010; Hadjipapas et al., 2007; Hall et al., 2005; Koelewijn et al., 2010; Muthukumaraswamy and Singh, 2008; Perry et al., 2013; Swettenham et al., 2009), paralleling neurophysiological findings in the macaque (Berens et al., 2008; Friedman-Hill et al., 2000; Frien et al., 2000; Gieselmann and Thiele, 2008; Gray and Singer, 1989; Henrie and Shapley, 2005). Although these stimulus-dependent effects in monkeys and humans were recorded in experiments that are difficult to compare, they suggest that gamma may be generated by neural mechanisms, which if not in common, are at least partially shared across the two species and across recording modalities. Yet, it is still not clear whether characteristics of gamma-band activity observed in invasive macaque recordings can be linked in a straightforward manner to characteristics observed from human non-invasive recordings. More specifically, can observed modulations in gamma frequency and power in a human MEG experiment be thought of as equivalent to frequency and power modulations in monkey electrophysiological recordings? This is a non-trivial question of crucial importance for non-invasive EEG/MEG studies on human brain functions that base their hypothesis on prior monkey electrophysiology studies.

Here we investigated whether the functional behavior of gamma oscillation frequency and power in the human is consistent with the behavior observed in the macaque and further whether it is a function of the scales of measurement (spikes, LFP, MEG). To this end, it was necessary to identify a stimulus parameter with robust and well-understood effects on gamma frequency and manipulate this parameter in both human and non-human primates. In computational studies, gamma is often modeled in terms of an interaction between excitation (E) and inhibition (I). In Pyramidal Interneuron Gamma (PING) networks (Tiesinga and Sejnowski, 2009), spikes from E-neurons quickly lead to spiking in I-neurons, which in turn briefly shut down the network. The recurrence of this process leads to oscillations, which are in the gamma range due to the time constants of the neurons involved (Börgers and Kopell, 2005). Increasing E-drive leads to a faster recovery from inhibition, resulting in a higher oscillation frequency (Brunel and Wang, 2003; Buia and Tiesinga, 2006; Tiesinga and Sejnowski, 2009; Traub et al., 1996). The gamma frequency increase predicted to follow from increased E-drive has been confirmed in neurophysiological studies in the macaque, in which gamma oscillation frequency as measured in the LFP recorded from early visual cortex was shown to increase strongly with stimulus contrast (Jia et al., 2013; Ray and Maunsell, 2010; Roberts et al., 2013). Recently contrast-induced frequency shifts have also been shown in human MEG (Perry et al., 2014). Moreover, PING models typically yield an optimal frequency at which gamma power is maximized, with lesser power for frequencies below or above the optimal frequency (Lowet et al., 2015; Roberts et al., 2013). Stimulus contrast can be thought of as an experimentally accessible proxy for E-drive to primary visual cortex (Contreras and Palmer, 2003; Sclar et al., 1990). Hence, parametric variation of contrast is a paradigm that is theoretically well understood such that it may permit the identification of biophysical mechanisms.

We applied this rationale to a human MEG experiment, which was designed for comparison with a previous study in macaque V1 conducted in our group (Roberts et al., 2013). Designing the MEG experiment for comparative purposes was not trivial: on the one hand, one needs to match the monkey and human experiments as closely as possible; at the same time due to the low gamma SNR in the MEG for a number of experimental conditions, some parameters in the MEG experiment need to be adapted such that sufficient SNR could be obtained. Despite the resulting limitations in achieving a perfect comparison, the study benefited from the use of a visual parameter in both studies of which the effects on gamma are well-understood, and from identical spectral analyses of human and macaque data. In addition, in the macaque data we also analyzed the gamma response in single-unit spike trains as a function of visual contrast. We found that increases in visual contrast led to similar shifts in gamma frequency in monkey single unit spiking, in monkey LFP, and in human MEG. Further, we found spectral asymmetries, which were consistently modulated by contrast in all recording modalities (single unit spikes, LFP, MEG). Alongside the similarities between MEG, LFP and spikes, we also observed a major difference: whereas there was a linear increase in the gamma oscillation power with stimulus contrast in the MEG data consistent with previous observations (Hall et al., 2005; Perry et al., 2014), in the LFP data, saturation and even reduction of power were observed at high contrasts.

Overall, the pronounced similarity of these findings across measurement scales and across human and macaque suggests major similarities in the generative model of gamma-band activity in macaque monkeys and humans. However, the pronounced difference in contrast-dependent power modulation indicates that MEG-measured gamma power cannot be easily equated to LFP gamma power. These results as well as theoretical considerations indicate that oscillation frequency might be a more robust parameter for comparing findings across species and spatial scales of measurement. At the same time it will be necessary to explore whether power differences are related to differences in gamma generating mechanisms between macaque monkeys and humans, or to potential differences in signals picked up during invasive neurophysiological recordings and MEG measurements.

Materials and methods

Human MEG recording

Participants

All 9 participants provided informed consent according to the guidelines of the local ethics committee (Commissie Mensgebonden Onderzoek/Committee on Research Involving Human Participants, Region Arnhem—Nijmegen, The Netherlands). The sample consisted of 4 male and 5 female participants. The age range was from 25 to 41 years and all were healthy and had normal or corrected-to-normal vision.

Task

Participants were positioned comfortably in the MEG scanner, and instructed to sit still while looking at a fixation spot in the middle of the translucent back-projection screen (an EIKI LCD projector that was outside of the shielded room, projection dimensions: 45 × 34 cm, W × H, refresh rate: 60 Hz), which was located 70 to 80 cm in front of them. The monitor was calibrated to linearize luminance as a function of RGB values.

Each experimental trial consisted of a 2 s interstimulus interval and a 2 s stimulation period. The fixation spot was present at all times. Stimuli were square-wave gratings of a 4° size positioned in the lower right visual quadrant (the left upper corner of the square was offset 0.15° from the fixation spot vertically and horizontally) presented on a gray background with the same mean luminance as the stimulus. Stimulus presentation was in MATLAB, using the Psychophysics Toolbox extensions

(Brainard, 1997; Kleiner et al., 2007; Pelli, 1997). Stimuli were static and had a spatial frequency of 3 cycles/degree. Stimulus luminance contrast was defined as Michelson contrast:

$$Cm = 100\,\frac{L_{\max} - L_{\min}}{L_{\max} + L_{\min}}$$

where $L_{\max}$ and $L_{\min}$ denote the maximal and minimal luminance of the stimuli. Stimulus contrast was varied between five values (20%, 36%, 48%, 66%, 96%) and selected pseudo-randomly for each trial (100 trials per contrast condition, per participant). After each block of 50 trials the participants had a small break of 10 s. To minimize long-term adaptation effects, orientation (0° or 90°) and phase (8 phases) of the grating was randomized trial-by-trial. To support the participants in keeping their gaze on the fixation point a change fixation task was implemented in which in rare occasions the fixation point changed slightly in color at the end of the stimulation period. The subject had to indicate the color change by a button press.

Recording

Magnetic fields of the brain were sampled at 1200 Hz by a 275 axial gradiometer whole-head magnetoencephalography system (Omega 2000; CTF Systems, Canada) in a magnetically shielded room. A third-order synthetic gradiometer was applied on the sensor data. The MEG signals were high (cutoff 0.5 Hz) and low pass filtered (cutoff 120 Hz) based on data segments of 10 s. The head position was localized with respect to the MEG sensors using fiducials (small coils) attached at anatomical landmarks (the nasion, the left and right ear canal). Head movements were continuously monitored during the MEG recording and kept below 0.5 cm. For anatomical source reconstruction of the MEG signal anatomical MRIs (T1-weighted, MPRAGE) were recorded using a 1.5 T whole-body scanner (Siemens Avanto, Erlangen, Germany) with anatomical reference markers (left and right ear canals) at the same locations as the MEG fiducials.

Beamforming

Synthetic Aperture Magnetometry, a minimum variance beamformer (Robinson and Vrba, 1999; Vrba and Robinson, 2001), was used to generate maps of power change on a 1 mm^3 grid throughout the brain for each participant. We compared the power in the 20–80 Hz band between the baseline (−1.8 s to 0 s) and stimulus (0.2 to 2 s) periods. Data from all luminance contrast conditions were used to generate the volumetric maps. For each participant, we identified the peak voxel showing the largest stimulus-induced change in power compared to the baseline, which was quantified as the pseudo-t parameter in Synthetic Aperture Magnetometry (Robinson and Vrba, 1999; Vrba and Robinson, 2001). We then reconstructed a time-varying estimate of the electrical activity for the peak location of the image for each participant (this was the estimated source signal). Briefly, this estimate was based on a spatial filter constructed from a covariance matrix comprising baseline and combined luminance contrast conditions. The covariance window for the source signal estimation was −2 s to 2 s, 20–80 Hz. All luminance contrast conditions were used in the construction of the spatial filter (the beamformer weights) and sensor data were then projected through this common spatial filter to yield a time-varying estimate at the image peak location (source signal) of each trial. All subsequent analyses were based on these estimated source signals.

Monkey electrophysiological recording

Details of the experimental methods in the monkey experiment have been described in Roberts et al. (2013).

Subjects and surgical procedure

Two adult male rhesus monkeys (*Macaca mulatta*) weighing 7 and 9 kg took part in this experiment. Two chambers were implanted above early visual cortex, one positioned over V1 and V2 and the second over V4. For the experiment reported here we used data from V1 only. A head post was implanted to head-fix the monkey during the experiment. All the procedures were in accordance with the European Council Directive 2010/63/EU, the Dutch 'experiments on animal acts' (1997) and approved by the Radboud University Ethical Committee on Experiments with Animals (Dier-Experimenten-Commissie, DEC).

Task

The monkeys were head-fixed and placed in a Faraday-isolated darkened box at a distance of 57 cm from a computer monitor (refresh rate 60 Hz; resolution 1280 × 1024 pixels). The monitor was calibrated to linearize luminance as function of RGB values. The position of one eye was monitored with an infrared eye-tracking system (Arrington, 60 Hz temporal resolution). The stimulus presentation, reward delivery and behavioral control were managed by the CORTEX software package (http://www.cnbc.cmu.edu/~rickr/ctxman5.html). Square wave gratings (2 cycles/degree, 3–5° diameter) were shown on an isoluminant gray background. Stimulus position was based on the mean receptive field position of the recorded neurons in V1 and V2 (4–6° eccentricity). Stimuli were presented at 8 different luminance contrasts (2%, 3.5%, 6%, 9.7%, 16.3%, 35.9%, 50.3%, 72%) and two orientations (45° and 135° in monkey S, 0° and 90° in monkey K) in pseudo-random order. Data from the two orientation conditions were pooled. The behavioral task was to hold fixation on a fixation spot in the middle of the computer screen during stimulus presentation. The within-trial median total range eye position was 0.3° in monkey K and 0.8° in monkey S. If the eye position exceeded a maximally 3 × 3° window centered on the fixation spot, the trial was aborted. Maintaining fixation was rewarded with a drop of fruit juice. A trial consisted of a 1 s pre-stimulus baseline, followed by a stimulus randomized in duration between 0.75 and 4 s. Trials with < 1.5 s stimulus duration were excluded because of insufficient frequency resolution.

Electrophysiological recording

V1 recordings were made with Plexon U-probes (Plexon Inc.) consisting of 8 carbon nanotube coated contacts (10 μm diameter, 0.5–1 mΩ impedance, and 200 μm inter-contact spacing). The probe was inserted through a sharp guide tube, which was lowered through granulation tissue just above the level of the dura surface. The probe was then advanced by a separate microdrive (Nan Instruments LTD.). The probe was connected to a headstage of high input impedance, and data were acquired via the Plexon 'Multichannel Acquisition system' (MAP, Plexon Inc.). The system employs two analogue hardware filters to extract two signals from the measured extracellular signal, namely the 'LFP signal' [0.7 Hz–300 Hz] and the 'spike signal' [150 Hz–8 kHz]. Next, the extracted analogue signals were amplified and digitized at two different rates, namely 1 kHz for the 'LFP signal' and 40 kHz for the 'spike signal', respectively. For analysis of LFPs, the quantifications obtained from each contact of a laminar probe were averaged over the 8 contacts of each laminar probe to obtain an overall estimate irrespective of the layering of visual cortex.

Spike sorting

Multi-unit spike channels were spike sorted using an adapted Waveclus approach (Quiroga et al., 2004). A k-mean clustering method was used, where the optimal cluster size was chosen by using the cluster size with the best ratio of between-cluster-distance to within-cluster-distance. The quality of the clustering was manually checked and in certain cases regrouping was performed.

Single unit spike trains

With this analysis we aimed to examine effects of contrast on gamma frequency at the single unit level. We used analysis of spike trains only, as opposed to the more commonly used spike-field coherence (SFC) measures, because our objective was to have a local measure of gamma periodicity in spikes, that was independent of LFP properties. We quantified spectral parameters (peak and mean gamma frequency and skewness index) that required a sufficiently strong gamma power induction in the spectrum for reliable estimation (see selection criterion below). However, most spike trains that we analyzed were only weakly oscillatory and did not display a gamma component. The reason for this is the sparseness of the neuronal firing, that is, most single neurons will not fire at each cycle but will rather skip cycles (see the Discussion section). Nonetheless, we were able to isolate a small number of neurons ($n = 5$ in monkey S and $n = 2$ in monkey K) that exhibited a strong oscillatory (gamma) component. Thus, our sample of neurons was not representative of the entire population of neurons but rather of the subpopulation of neurons that fire on many cycles of the oscillation. Here we tested the hypothesis of whether these cells exhibit the same modulations as a function of contrast as those observed in the LFP and MEG.

Data analysis common for human MEG and monkey microelectrode data

Spectral analysis

The FIELDTRIP MATLAB toolbox (http://www.ru.nl/neuroimaging/fieldtrip) (Oostenveld et al., 2011) was used for spectral analysis. For the human MEG power spectra time windows of [−1.8 s to 0 s] as baseline and [0.2 s to 2 s] as stimulus periods were used. For the monkey power spectra time windows [−1 s to 0 s] as baseline and [0.2 s 1.5 s] as stimulus periods were used. A minimum trial duration of 1.5 s for the monkey data was chosen because, although stimuli were presented for up to 4 s, selecting only trials with maintained fixation for longer durations would lead to a strong decline in the number of trials available for analysis. The FFT power spectra were computed using a multitaper method using discrete prolate spheroid sequences (DPSS) with a spectral smoothing of ±3 Hz. For the time–frequency analysis (TFR), we also used FFT DPSS computed on sliding windows of 0.25 s length with 10 ms spacing. Prior to computation of any spectral parameters we calculated the power ratio between the stimulus period and the baseline period. Thus subsequent calculations were based on such baseline-corrected spectra. The power ratio was defined as:

$$\text{Power ratio} = \frac{(\text{stim}-\text{base})}{\text{base}}$$

where stim is the stimulation period (presentation of the visual stimulus) and base is the baseline period (absence of the visual stimulus).

Asymmetric Gaussian fitting for spectral parameter estimation

To estimate the peak frequency, power and asymmetry we used a fitting approach, where the two sides of a Gaussian were separately fitted.

$$\lambda(x) = \lambda_0 e^{-x^2/2\sigma^2} \begin{cases} \sigma = \sigma L, \text{if } x<0 \\ \sigma = \sigma R, \text{if } x \geq 0 \end{cases}$$

where x represents the frequency of the spectrum, λ_0 is the amplitude of the Gaussian and σL and σR are the left and right standard deviations respectively. The fitting was implemented using an unconstrained nonlinear minimization algorithm (fminsearch, Matlab, MathWorks). Before Gaussian fitting the spectra of a subject were normalized to 1 (over all contrasts). The frequency range was from 25 to 70 Hz in human MEG and 20 to 60 Hz in monkey data.

Gamma power

Gamma power was estimated as the area under the curve of the fitted Gaussian. This measure takes changes in the bandwidth into account in comparison to peak power measures. Certain contrast conditions might lead to stronger moment-by-moment frequency fluctuations leading to broader gamma power distributions. Taking the area under the curve ensured that power estimates were not confounded by changes in the gamma band bandwidth.

Gamma frequency

Gamma frequency was estimated by two methods. We computed the peak power frequency (mode) or the center of weight (mean) frequency. They were estimated based on the fitted Gaussian function: For each power spectrum, we fitted the asymmetric Gaussian using the nonlinear minimization algorithm. We then derived the mode as well as the mean of the fitted asymmetric Gaussian. This gave us reliable estimates for human MEG as well as for monkey LFP/spike data. We could use the mode and mean estimates of the Gaussian to compute an asymmetry index. This is because distribution asymmetry affects the mode and the mean differently. Moreover, the mean frequency compared to the more commonly used peak power frequency is a more robust measure since it is less affected by noise in the power estimate, which can strongly affect peak power frequency (small variations on a broader gamma peak can lead to strong variations in peak power frequency estimates).

Distribution asymmetry

Distribution asymmetry was quantified by an index of skewness (Doane and Seward, 2011)

$$\rho = (\mu - \kappa)/\sigma$$

where ρ is the nonparametric Pearson's first skewness coefficient, μ the mean, κ the mode and σ the standard deviation of the distribution. To approximate σ we used the standard deviation of the fitted Gaussians.

Statistical analysis

Analysis of spectral power in Fig. 2 included data from a single session in 9 human participants (100 trials per contrast). Analysis of spectral power in Fig. 3 included data from 15 sessions from monkey S (59.0 ± 2.7 trials per session and contrast) and 13 sessions from monkey K (42.8 ± 1.2 trials per session and contrast).

Regression analyses of spectral parameters

Regression analyses of *power* were run on all available contrast conditions and sessions. However, the accurate estimation of spectral parameters such as *peak/mean frequency* and *spectral asymmetry* requires sufficient power induction. Therefore, when quantifying peak frequency and spectral asymmetry, analysis was restricted to data showing a clear oscillatory gamma component. To this end, only monkey sessions that showed significant gamma power induction (if t-test p-value (25–60 Hz) <0.05) in at least three contrasts were included in the analysis; sessions not satisfying this criterion were excluded as a whole.

The absence or weakness of induced gamma responses in some experimental sessions in the monkey was due either to an insufficient number of trials or due to the positioning of the laminar probes largely in the deep cortical layers where gamma is weak (Roberts et al., 2013; Xing et al., 2012). Data then included 11 sessions from monkey S (average of 700.1 trials per contrast pooled over all sessions) and 11 sessions from monkey K (average of 470.5 trials per contrast pooled over all sessions). The same selection rule (at least 3 contrasts with significant gamma power) was applied to the human population at the participant level, leading to a selection of 7 of 9 subjects (who each contributed a single session with 100 trials per contrast). Within this data selection, we excluded the lowest contrast for MEG frequency and

asymmetry estimations (20%), and the three lowest contrasts for LFP/spike-based frequency and asymmetry estimations (2%, 3.5%, 6%), because of a lack of gamma power (t-test p-value [25–60 Hz] >0.05). Hence, in these cases, monkey recording analysis was restricted to contrasts 9.7%, 16.3%, 35.9%, 50.3% and 72%, and human MEG analysis to 36%, 48%, 66% and 96%.

Statistical regression analysis was done on spectral parameters of interest over contrast on a single trial level basis for each subject individually (human participant or monkey). To accomplish this, we used a robust fitting approach (reducing the effects of outliers) to assess significance and effect size estimates on the level of individual subjects (monkey or human participants) or sessions (in monkey data). We also applied regression fitting on the trial-averaged spectra of all subjects to estimate population effects. Note that the total pooled sample of trials available per individual contrast from the two monkeys (1170.6) was comparable to the total pooled sample of trials available per individual contrast from the 9 human participants (900). In the analysis of frequency and asymmetry, analysis was restricted to 7 participants (700 trials per contrast, see above). In the regression analyses we fitted both linear and quadratic models in a stepwise fashion — quadratic fits are only reported if they substantially improved the explained variance (R^2) compared to linear models.

Results

Comparison of stimuli and experimental designs

In a previous study from our group, Roberts et al. (2013) reported a ~25 Hz increase in LFP frequency in V1 and V2 of 2 macaque monkeys as luminance contrast was increased in square-wave gratings from 2% to 72% (grating diameter 3–5°, eccentricity 4–6°, spatial frequency 2 cycles/degree., lower visual field quadrant, 1300 ms analysis time window). To test the generality of the relationship between grating contrast and gamma frequency, we compared this monkey data with MEG data collected from 9 human participants also viewing square wave gratings. Luminance contrast was increased from 20% to 96% (grating diameter 4°, eccentricity 3°, spatial frequency 3 cycles/degree, lower visual field quadrant, 1800 ms analysis time window). In both experiments (Figs. 1A–B), the subjects' task was to fixate on a central fixation spot; the grating stimuli were irrelevant to the observer. MEG provides a less direct measurement of activity than invasive measurements of LFPs and spiking (Fig. 1C). We chose the stimulus conditions in the MEG experiments to be as close as possible to those used in the monkey recording experiment, while not losing too much signal-to-noise. This led to the choice of a slightly smaller eccentricity and a consequent choice to present the grating in a square aperture, rather than a circular one, to limit the stimulus to a single quadrant. The lower signal-to-noise of the MEG signal also required us to shift the contrast manipulation to higher contrast values. Despite the large but necessary differences in contrast range, and some other smaller differences in stimulus parameters outlined above, the overall similarity of the experimental manipulation permitted a good comparison of the spectral properties of the MEG signals (top panels in Figs. 1D, E) with the LFP signals (middle panels in Figs. 1D, E), and the spiking signals (bottom panels in Figs. 1D, E).

Spectral properties of gamma MEG data in human visual cortex

In 8 of the 9 participants, the peak of the MEG beamformer image was found in the visual cortex contralateral to the stimulus presentation, consistent with many previous studies using a similar stimulation (Duncan et al., 2010; Hall et al., 2005; Muthukumaraswamy et al., 2010; Swettenham et al., 2009). The mean pseudo-t value (over all contrasts) was 4.5 (SD = 2.8). We note that in participant 2 (Fig. 2, middle panel, top row) the gamma SNR was very low. The weights of the beamformer for the peak voxel were extracted and used for the reconstruction of the source time series (see the Materials and methods section). All analyses performed were based on this source reconstructed signal, which beyond a localization to V1 also offers a higher SNR compared to the sensor data (Brookes et al., 2009; Hadjipapas et al., 2007).

Fig. 2 shows the effect of contrast on the power spectra obtained in the MEG experiment. In Fig. 2A, trial-averaged Time–Frequency Representations (TFRs) from a sample participant (participant 8, middle panel, bottom row, Fig. 2B) exhibiting a strong gamma response are shown for the full range of contrasts used. A gamma band was discernible at stimulus contrasts of 36% and higher (Fig. 2A). For these contrasts, there was a transient power increase in the gamma range briefly after stimulus onset, followed by weaker power as long as the stimulus remained in view. In addition, gradual frequency decay is observed across the trial. These observations are consistent with previous observations of the MEG response in the human (Hadjipapas et al., 2007; Muthukumaraswamy et al., 2010; Swettenham et al., 2009). Fig. 2A also shows that both relative gamma power and gamma peak frequency increased with contrast. Fig. 2B shows the spectra for each participant. As illustrated in Fig. 1E, each power spectrum represents the relative power change ratio between a spectrum corresponding to the stimulus period and a spectrum corresponding to the baseline period; spectra were obtained using Fourier analysis and DPSS tapering (see the Materials and methods section for details). Despite individual variation, these data overall are in line with the average power increase and the shift of the gamma band towards higher frequencies with increasing stimulus contrast, as shown in the individual participant in Fig. 2A. However, some participants showed very weak gamma power across all contrasts. Of particular interest in Figs. 2A and B was the peak–power frequency of the gamma spectrum, also referred to as '*peak frequency*'. When averaging the spectral data over participants, the effect of power visually dominates in the figure (Fig. 2C, top), although the frequency shift of the spectra is still evident. To better visualize the peak frequency shift, we normalized all spectra to the range of [0, 1] for each contrast (Fig. 2C, bottom). From 36% to 96% contrast, the peak frequency averaged across participants increased by ~8 Hz (see quantifications below).

Gamma spectral properties of neurophysiological data in macaque V1

For comparison with the human MEG data, Fig. 3 shows electrophysiological data from area V1 of the two macaque monkeys. Fig. 3A shows the TFR of a single sample session from monkey S (59 trials per contrast on average). There was observable gamma power at a contrast as low as 6%, with a frequency of ~28 Hz. With increasing contrast, the peak frequency of sustained gamma increased to 45 Hz at 72% contrast. Figs. 3B and E show the LFP gamma spectra for monkey S (15 sessions with on average 59 trials per session and contrast) and monkey K (13 sessions with on average 43 trials per session and contrast). The spectral data were averaged over the whole laminar probe (see the Materials and methods section). The LFP spectra showed a large shift in frequency as a function of contrast, as well as a large modulation in power. Figs. 3C and F show the contrast–response function for the two monkeys based on population spike rate for all sessions combined, normalized by baseline (120 and 104 multi-unit spike channels in monkey S and K respectively). Spike rate increased as a function of contrast, reaching asymptotic levels at the highest contrasts, in agreement with previous observations (Albrecht and Hamilton, 1982; Sclar et al., 1990). Spike train spectral analysis in Figs. 3D and G was based on a Fourier transform applied directly to the spike trains of a total of 7 single well-isolated (spike-sorted) neurons (5 neurons in monkey S, 2 in monkey K). By working on the spike trains only rather than using Spike Field Coherence (SFC) we ensured that we have a measure of single unit rhythmicity, as spikes are derived independently from the LFP. The low number of neurons used for spike spectral power analysis was due to the requirement of a strong spectral gamma

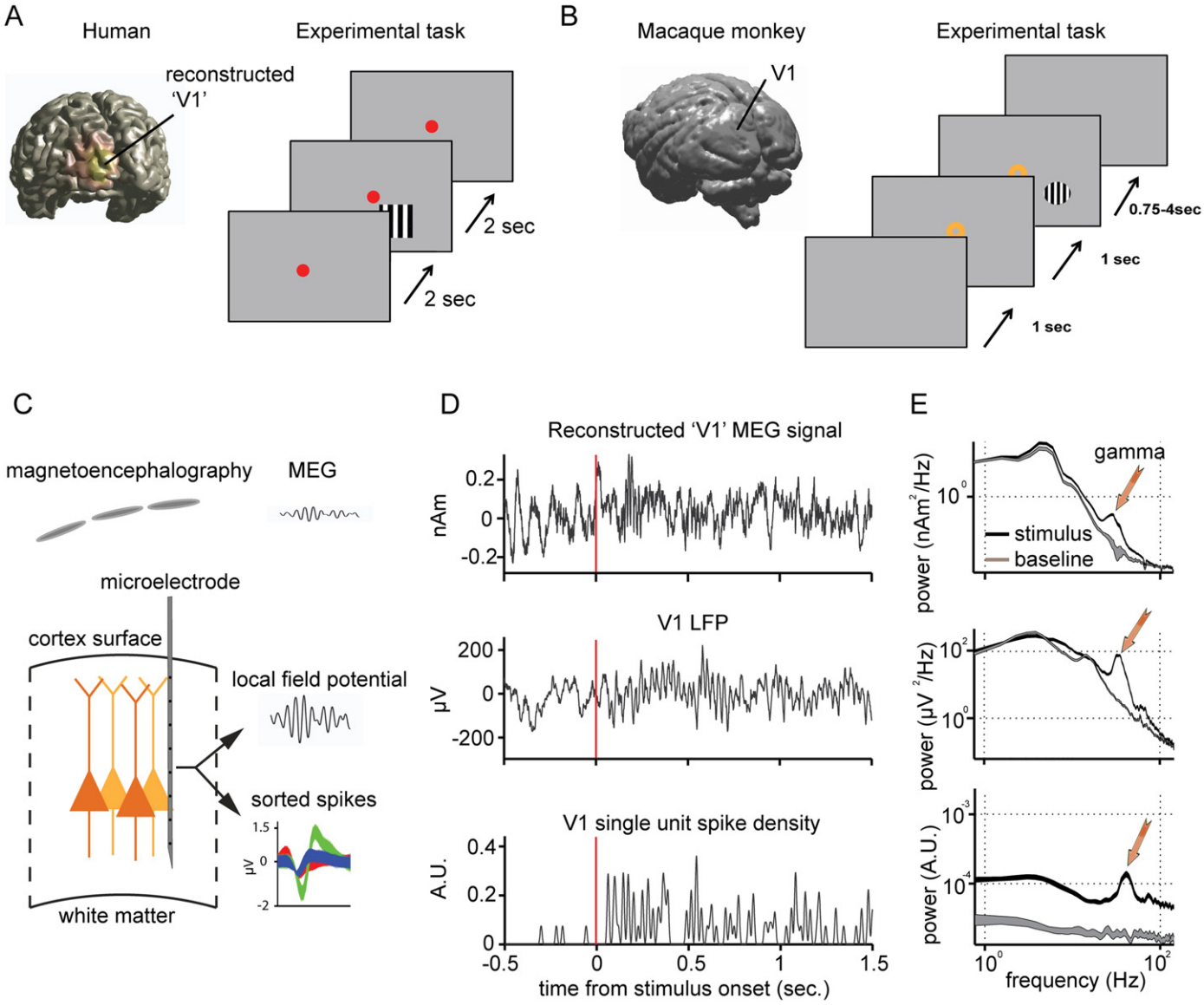

Fig. 1. Overview of experimental tasks and recording signals. A) Left: A beamformer-reconstructed gamma source projected on an MRI-reconstructed human brain. The peak voxel (located in the area of highest activation as indicated by the yellow color) for each subject was chosen for source signal reconstruction using the beamformer weights for that location. Right: human experimental paradigm. B) Left: Illustration of a macaque brain. Right: monkey experimental paradigm. C) Illustration of the three signal sources compared in this study: MEG, LFP and extracellular single unit activity. MEG collects magnetic signals (fT-pT) over larger cortical space from outside the brain; it is sensitive to highly synchronized events representing mostly dendro-somatic currents (>5 mm). The LFP (microV) is predominantly sensitive to dendro-somatic currents locally (<0.5 mm to the electrode). Extracellular spikes represent action potentials. After spike sorting, a specific set of spikes can be attributed to one cell. D) Examples of single trial data are shown for the three different signal types. The spike train was converted into spike density to facilitate the observation of the gamma rhythm. E) The log–log power spectra of a baseline compared to high contrast grating stimulation (single subject/session). A specific increase in the gamma band can be observed. There were considerable differences in power induction between MEG, LFP and spikes.

component (see the Materials and methods section). Despite the fact that the spike train spectral properties were derived independently from the LFP, the results from spike spectral power analysis resembled those from LFP analysis, especially in monkey S. In both the LFP and spiking spectral analysis, monkey S showed a lack of saturation of the frequency shift at high contrasts, and a decrease in gamma power for contrasts exceeding 35.9%. Comparing LFP and spiking analysis in monkey K shows a similar pattern of gamma frequency increases and clear saturation of gamma power at higher contrasts (with a decrease in spike-based spectral power at the highest contrast). Hence, within monkeys there was close similarity in LFP and single unit spiking spectral analysis (in Fig. 3 compare B, D and E, G), but both analyses showed consistent differences between monkeys (in Fig. 3 compare B, E and D, G). Despite these differences in spectral data, spike rate as a function of contrast was similar between the two monkeys (in Fig. 3 compare C, F). In addition, it is worth noting that whereas spike rates in both monkeys (Fig. 3C and Fig. 3F) increased monotonically with contrast, this was not the case with the LFP gamma power (Fig. 3B and Fig. 3E). In monkey S, spike rates (Fig. 3C) continued to go up at high contrasts despite a concurrent decay in LFP gamma power (Fig. 3B). In monkey K, while firing rates also increased at high contrasts (Fig. 3F) LFP gamma power saturated (Fig. 3E). That is, dissociation between firing rate and LFP gamma power is observed (Musall et al., 2014).

Comparison of contrast-induced changes in gamma spectra between spikes and LFPs in the macaque and human MEG

In Fig. 4, we summarize the effects of luminance contrast on the spectral parameters for the three signal domains (MEG, LFP and spikes). Figs. 4A–C show, respectively, the peak frequency in the MEG, the LFP, and in the periodic component of single unit spike trains. Figs. 4D–F

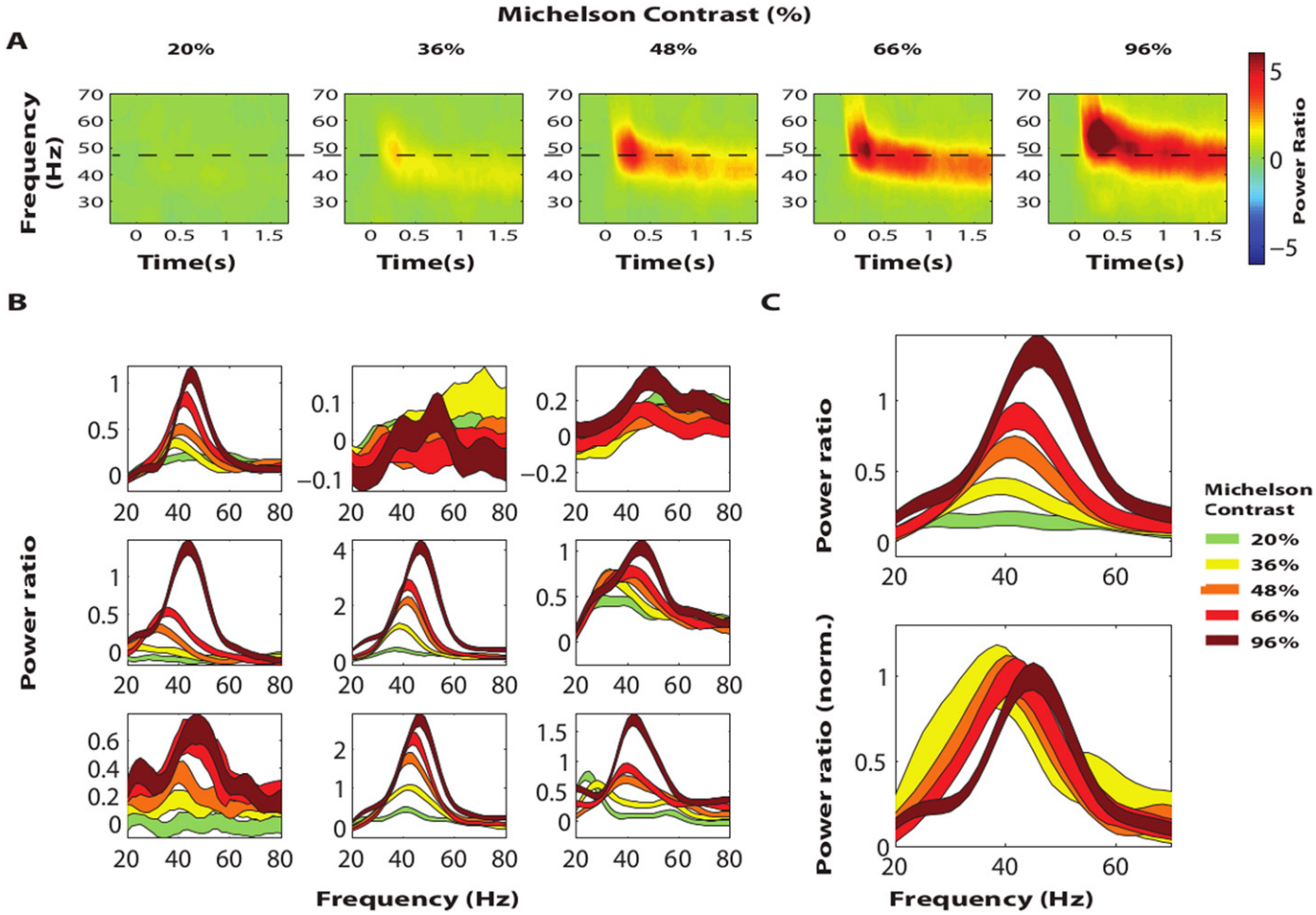

Fig. 2. Human MEG results. A) Time frequency representation (TFR, time window for baseline correction −2 to 0 s) for different stimulus contrast conditions of participant 8. B) The power spectra (relative power) for the individual participants (1 to 9 ordered from the left upper to the right lower panel). Error bars represent ±1 standard error of the mean. Participants had different relative power levels (Y axis values). C) Average of individual power spectra before (top) and after normalization per contrast of power spectra of the individual participants.

show the power ratio relative to baseline. Figs. 4A and D represent human MEG signal, Figs. 4B and E show monkey LFP, and Figs. 4C and F show monkey single unit spike trains, as a function of contrast. Notice that the contrasts included differ in the different figure panels (for details, see the Materials and methods section, the Data analysis common for human MEG and monkey microelectrode data section, and next two paragraphs). In the monkey LFP data, the peak frequency shifted from ~26 Hz at 9.7% contrast to ~45 Hz at 72% contrast, hence a shift of ~19 Hz. In human MEG, the shift from 32% contrast to 96% contrast was ~8 Hz. This is in good agreement with the 7.6 Hz shift in peak frequency observed in monkeys for the most comparable contrast variation from 36% to 72% contrast. The frequency of the gamma periodic component of the spike trains estimated based on the 7 single units (pooled over monkeys, from Figs. 3D and G) increased with contrast over a similar frequency range as the LFP peak frequency. Note that, contrary to power, the relationship between contrast and frequency was monotonic across all recording modalities, species and spatial scales (spikes, LFP, MEG). Figs. 4D–F show contrast-induced shifts in power ratios for the MEG, LFP and spiking signals. Interestingly, despite some differences between the monkeys (Fig. 3), the average spectral power of monkey LFP and spiking signals saturated at high contrasts, while average MEG power showed a nearly linear relation over the whole contrast range. The latter observation is in line with the individual subject data in Fig. 2B and with previous MEG studies (Hall et al., 2005; Perry et al., 2014). The main observations in each panel in Fig. 4 were tested statistically by regression analyses, which are described below.

Across-species effects of contrast on gamma peak frequency

We first statistically tested the relationship between peak frequency and contrast in the human participant data on the data shown in Fig. 4A. However, note that peak frequency could not be estimated when the gamma spectrum did not differ significantly from baseline (if t-test p-value (25–60 Hz) >0.05). Participants in which this was the case for >3 contrast conditions (2 of 9 participants) were excluded from the individual analysis of spectral power and peak–power frequency (participants 2 and 3, see Fig. 2). In the 7 remaining participants, we also excluded the lowest contrast from analysis as the stimulus-induced gamma spectrum did not differ significantly from the baseline, leaving 4 contrasts for analysis. In these 7 participants, we estimated peak frequency based on a fitted asymmetric Gaussian at the single trial level. The relationship between *single-trial* estimates of peak frequency and contrast was well fitted by a linear regression in each participant (all $p < 0.05$, average $R^2 = 0.072$, range R^2: 0.09 to 0.264). Using a single-trial approach, the explained variance was however only a fraction of the total trial-to-trial frequency variation. A part of the unexplained variance was likely accounted for by non-physiological noise inherent to the MEG measurements. In addition, gamma frequency is known to vary widely from moment to moment and from trial to trial even under constant conditions in the monkey (Brunet et al., 2013, 2014; Burns et al., 2011; Roberts et al., 2013). Most of this unexplained variability in the regression analysis was removed when *averaging* the single trial estimates within contrast conditions per participant (SE bars in Fig. 4A correspond to this analysis). Using the 4 (contrasts) × 7 (participants) averages for the regression, the linear relationship of

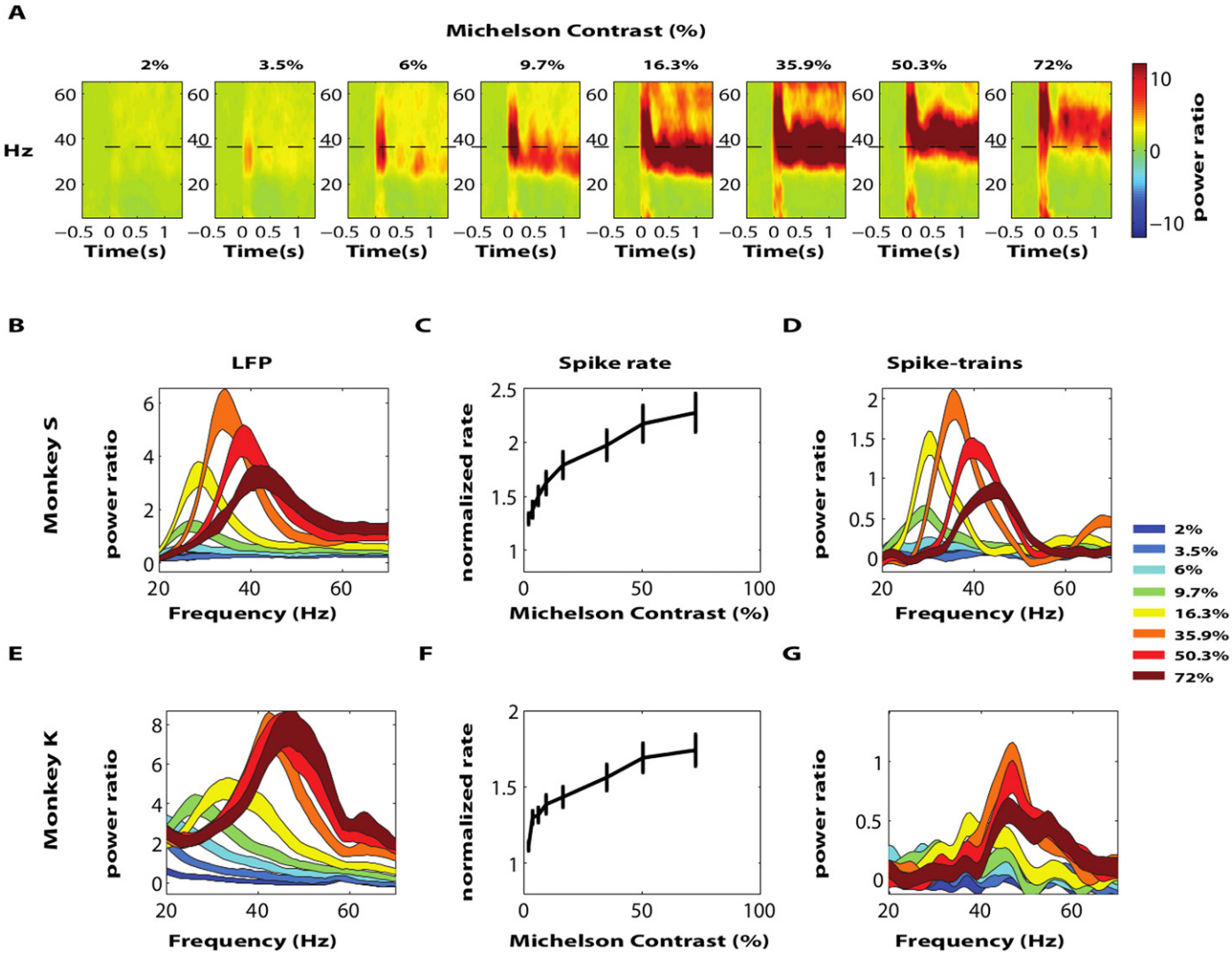

Fig. 3. Results from monkey V1 recordings. A) TFR from a sample session from monkey S for eight different contrasts (time window for baseline correction −1–0 s). B, E) LFP relative power spectra for monkeys K and S. Error bars represent ±1 standard error of the mean. Contrast modulated total power, peak frequency and asymmetry of the spectra. C, F) Mean spike rate is shown as a function of contrast for monkeys S (120 multi units) and K (104 multi-units), respectively. In both monkeys average spike rate increased with a stronger slope at lower contrast ranges. D, G) The power spectra of single neuron spike trains for the different contrast conditions in monkey S (5 neurons) and monkey K (2 neurons). The spectra for these spiking data were similar to the LFP power spectra.

the peak frequency of the spectrum to luminance contrast was again highly significant in human MEG but this time with much higher explained variance ($F = 29.7$, $R^2 = 0.534$, $df = 26$, $p < 10^{-5}$). There was no substantial increase in R^2 when comparing a quadratic to a linear fit. The regression slope corresponded to an increase of 0.16 Hz per % contrast.

We then statistically tested the relationship between peak frequency and contrast in the two monkeys on data shown in Fig. 4B. Sessions that had no significant gamma power induction in at least three contrasts were excluded (if t-test p-value (25–60 Hz) >0.05), leaving 11 sessions in monkey S and 11 sessions in monkey K (for details, see the Materials and methods section). Because of low power in the lowest contrasts used, the contrast–frequency relationship was tested only over the 5 highest contrasts in the monkey data (Fig. 4B). In a *single trial* linear regression analysis of LFP in the 22 available sessions, we found a robust relationship between peak frequency and contrast in each session estimate in both monkey S LFP (R^2(mean) = 0.452, all $p < 10^{-3}$) and monkey K (R^2(mean) = 0.448, all $p < 10^{-5}$). We then entered the *averages* per contrast and per session into the regression analysis (sample of 22 sessions × 5 contrasts). We found a strong linear relationship for the influence of contrast on the LFP frequency ($F = 159$, $R^2 = 0.63$, $df = 112$, $p < 10^{-5}$). The regression slope over the whole contrast range corresponded to an increase of 0.254 Hz (over both monkeys) per % contrast.

We continued by testing the relationship between gamma frequency derived from single neuron spike trains and contrast (Fig. 4C) in 7 well isolated single units collected in the same contrast conditions and sessions as used for LFP regression analysis. For the spiking data we tested the linear regression immediately on the averages per neuron and per contrast (7 neurons × 5 contrasts), yielding an effect size comparable to that found for the LFP data ($F = 47.8$, $R^2 = 0.61$, $df = 32$, $p < 10^{-5}$). The regression slope over the whole contrast range corresponded to an increase of 0.192 Hz (over both monkeys) per % contrast.

Overall we observed a striking similarity of the contrast frequency regression slopes between the estimates obtained at the MEG, LFP and single-unit spike trains. Fig. 5 depicts the regression slopes for individual MEG participants, LFP monkey subjects and single-unit spike trains

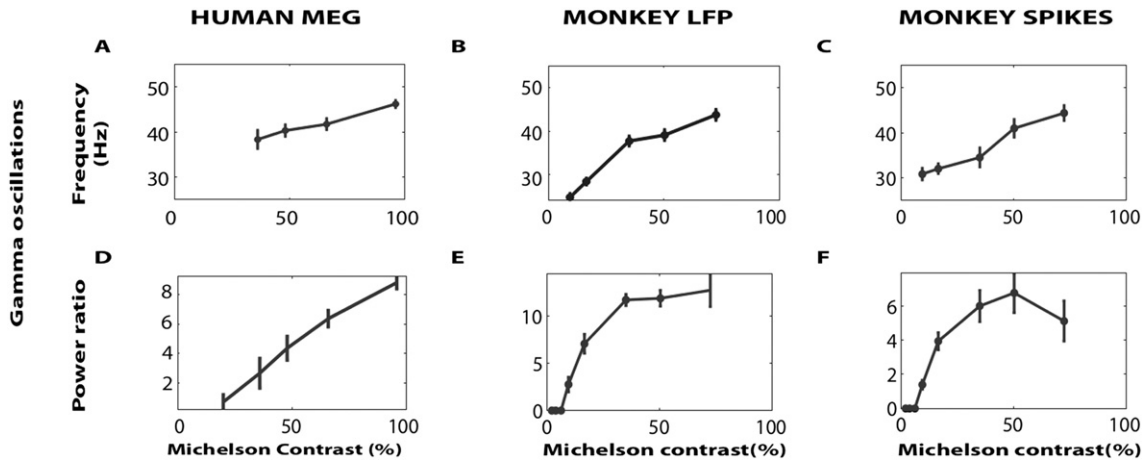

Fig. 4. Quantification of gamma spectral effects in humans and monkeys. A–C) The gamma frequency plotted as a function of contrast for human MEG data (A), monkey LFP data (B) and monkey spiking data (C). D–F) Gamma power plotted as a function of contrast for human MEG data (D), Monkey LFP data (E) and monkey spiking data (F). Monkey neurophysiological data are analyzed for sessions that showed significant gamma signal and likewise human MEG data are analyzed in participants with significant changes of gamma power from baseline (7 out of 9) (details in the text). Means represent the mean of session averages in the monkey neurophysiology data (22 sessions total), and of participant averages (N = 7) in the human MEG data. Error bars represent ±1 standard error of the mean.

also averaged by monkey subject. Given the very different spatial scales of gamma observation, a remarkable agreement is observed in terms of the estimated contrast–frequency gradients.

Across species effects of luminance contrast on gamma power

We found a strong modulation of gamma power (estimated by the area under the fitted Gaussian) as a function of contrast. In human MEG (Fig. 4D), linear regression fits on *single trial* estimates were performed in each of the 7 subjects also included in the analysis of frequency, but now also including the lowest contrast. In all participants, linear regression yielded significant results ($p < 0.05$; mean $R^2 = 0.11$, range $R^2 = 0.003$ to 0.33). Analysis on data *averaged* per contrast (5) and per participant (7) yielded a robust linear relationship ($F = 32.9$, $R^2 = 0.56$, $df = 26$, $p < 10^{-5}$).

For monkey LFP power (Fig. 4E), we first performed a regression analysis based *on single trials* on the 11 sessions from each monkey used in the previous section for analysis of frequency, but now including all 8 contrasts in the analysis. We found a significant linear relationship, but also found that adding a quadratic term improved the fit of the gamma power versus contrast relationship. This held in monkey S (all $p < 10^{-5}$, $R^2(\text{mean}) = 0.16$, $\Delta R^2(\text{linear} \rightarrow \text{quadratic}) = 0.12$) and monkey K (all $p < 10^{-5}$ $R^2(\text{mean}) = 0.155$, $\Delta R^2(\text{linear} \rightarrow \text{quadratic}) = 0.027$). Quadratic fitting on the sample of data averaged at the session level for 22 sessions × 8 contrasts gave a good fit with a substantial improvement against a linear fit ($F = 68$, $R^2 = 0.58$, $df = 112$, $p < 10^{-5}$, $\Delta R^2(\text{linear} \rightarrow \text{quadratic}) = 0.2$). We also performed quadratic regression on the power of the periodic component of single unit spike trains (data averaged for 7 neurons and 8 contrasts, Fig. 4F) and found a good quadratic fit, which was superior to a linear fit ($F = 8.67$, $R^2 = 0.35$, $df = 31$, $p = 1.1 \times 10^{-3}$, $\Delta R^2(\text{linear} \rightarrow \text{quadratic}) = 0.344$). Therefore, the analyses of power in Figs. 4D–F confirmed the monotonic linear relationship between contrast and power in human MEG and the nonlinear relationship between contrast and power in monkey LFP and single unit spike train data.

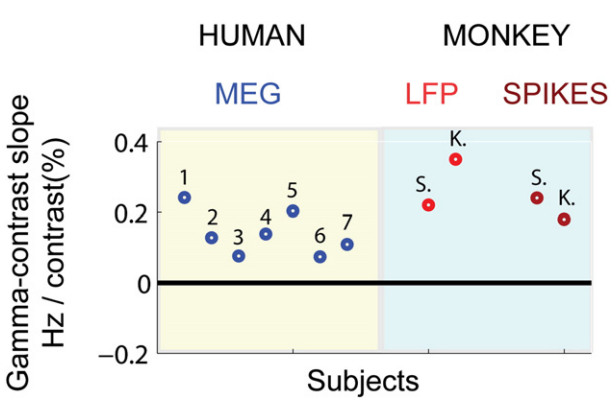

Fig. 5. Contrast–gamma frequency regression slopes at different spatial scales. The regression slopes between the frequency of spectral gamma peak as a function of luminance contrast for the 7 human subjects with a significant gamma power increase from baseline (details in the text) and the two monkeys were computed. For robustness considerations, the slope fitting was based on trial-averaged estimates. For the human subjects the slopes were quantified based on the MEG power spectra, whereas for the monkeys quantifications were based on the LFP and spike power spectra.

Effects of luminance contrast on the asymmetry (skewness) of the spectral gamma distribution

Beyond the effect on the power and the peak frequency of the spectra, contrast also had an effect on the shape of the spectra which could be expressed in terms of spectral asymmetry. In Fig. 6, the top row shows the spectral distributions for examples of lower and higher contrasts for MEG (Fig. 6A), LFP (Fig. 6B) and single unit spike trains (Fig. 6C), using averages of all available trials from respectively all participants, monkeys, or neurons. Considering the monkey LFP and spiking data first (Figs. 6B, C), the spectral distribution for a 16% contrast stimulus (red) was skewed to the right, while the spectral distribution for a 50.3% contrast stimulus (blue) appeared more symmetric. This change in skewness from lower to higher contrasts was weaker in the average MEG data (Fig. 6A), but visible in single subjects (Fig. 2B). The skewness of the spectral distribution was assessed by fitting an asymmetric Gaussian, and then comparing the fit's peak frequency (mode) with the center of weight frequency (mean), using a skewness index (see the Materials and methods section for details). When the mean is larger than the peak frequency, the distribution is skewed to the right (positive skewness index), and when the reverse is true it is skewed to the left (negative skewness index). Figs. 6D–F show the changes in skewness as a function of contrast in human MEG data, as well as monkey LFP and single unit spiking data on the same datasets as used for the analysis of contrast-induced shifts in peak frequency. In MEG, LFP, and spiking data, skewness evolved from positive (skewed to the right) to zero (symmetrical) and even slightly negative at the highest contrasts (skewed to the left). The change in skewness with contrast was quantified by linear regression on MEG data averaged at the participant level

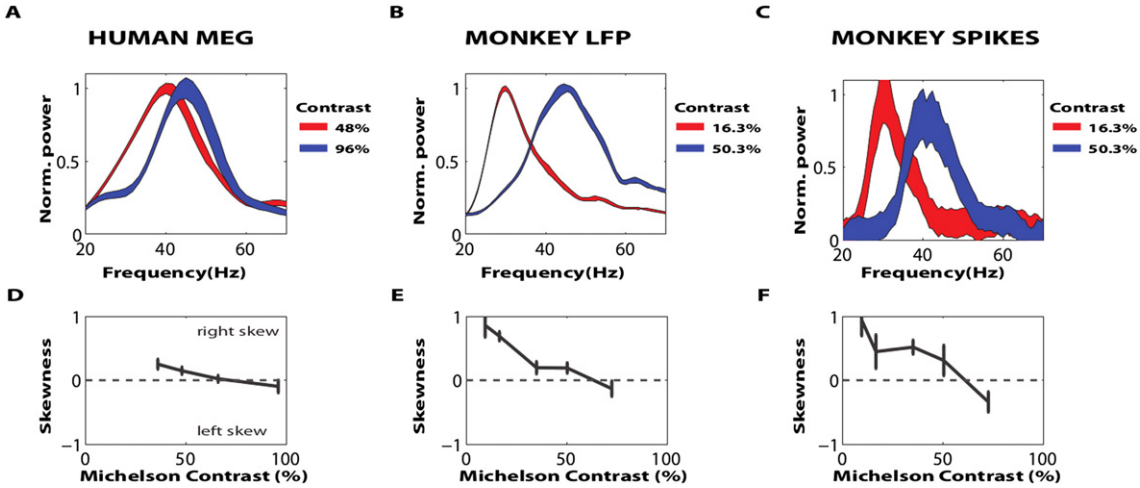

Fig. 6. Spectral asymmetry modulated by contrast. The three columns represent MEG, LFP and spikes respectively. The top row (A–C) depicts two relative power spectra of lower (red) and higher (blue) contrast in the different modalities. The spectra were normalized to [0 1] for each contrast to emphasize shape rather than power differences. Note the skewness to the right at lower contrasts and the relative symmetry at higher contrasts. The bottom row (D–F) shows the quantification of the spectral skewness. Error bars represent ±1 standard error of the mean. In all modalities the right-sided skewness decreased with contrast; in the case of spikes and MEG at high contrast there is even a reversal to left-sided skewness. The same data samples are used as in Fig. 4.

(N = 7) (F = 16.8, $R^2 = 0.4$, df = 26, $p = 0.4 \times 10^{-3}$), LFP data averaged at the session level pooled over monkeys (N = 22) (F = 21, $R^2 = 0.17$, df = 113, $p = 0.1 \times 10^{-3}$) and single unit spiking data averaged at the neuron level (N = 7) (F = 8.3, $R^2 = 0.22$, df = 32, $p = 0.72 \times 10^{-2}$). In summary, independent of data modality, an increase in contrast had three effects on gamma spectra, namely an increase in peak frequency, an increase in power (which however in monkeys showed saturation), and a decrease in right-sided skewness.

Discussion

In this paper, we present human MEG and monkey electrophysiological (LFP and spiking) recordings obtained in V1, where gamma oscillations were studied under similar stimulation parameters. MEG beamformer source reconstructions are well-suited to make this comparison due to their capacity to localize sources to visual cortex and to improve signal SNR (Brookes et al., 2009; Duncan et al., 2010; Hadjipapas et al., 2007). We used a paradigm where luminance contrast is varied parametrically. Because contrast is a proxy for afferent excitatory drive to visual cortex, this leads to a theoretical prediction about changes in gamma oscillation frequency. Gamma frequency is a fundamental network parameter that affects both internal structuring of units within the network and also longer-distance communication between populations (Bosman et al., 2012; Fries, 2009; Fries et al., 2007; Gregoriou et al., 2009; Roberts et al., 2013). The approach illustrated here permits testing the extent to which network mechanisms underlying gamma in non-human primates can be extended to humans. We report three main findings: First that in human MEG data gamma frequency increased over a substantial range as a function of luminance contrast, both on the population and single subject levels. This replicates and extends recent findings (Perry et al., 2014), and fits well with our monkey LFP and single unit spike data. Second, the MEG data showed a monotonic contrast-dependent increase in gamma power consistent with previous MEG studies (Hall et al., 2005; Perry et al., 2014), which contrasted with the power saturation and decay observed in our macaque V1 data at higher contrasts. Third, we present the first experimental evidence for contrast-induced modulations of spectral asymmetry, observed in human MEG as well as monkey LFP and single unit spikes. Below, we will discuss the implications of the three main observations, especially with respect to the extent to which these constitute support for common gamma mechanisms between the two species and for generalization of gamma spectral features across the different measurement scales (single neuron spikes, LFP and MEG signals).

Frequency modulation of MEG and LFP gamma oscillations as well as single unit spikes by stimulus contrast

In the MEG participants there was a systematic, approximately linear change in the dominant gamma oscillation frequency with stimulus contrast over the range of contrasts where estimations were possible. The frequency shift was significant in 7 out of 9 participants, in which the 2 participants with no significant effect also had very low induced gamma power, limiting frequency estimations. All participants with significant gamma power induction had a frequency change in the same direction. The frequency modulations were very similar to the one observed in the LFP of our rhesus macaque experiments and in other studies in the macaque (Jia et al., 2013; Ray and Maunsell, 2010) and also to a recent study of human MEG (Perry et al., 2014).

We aimed to test whether results found in population signals could be extended to behavior of single neurons. To this end we analyzed spike trains in isolation, as opposed to the more commonly used spike-field coherence (SFC) measures, in order to have a local measure of gamma periodicity in spikes independent of LFP properties. We quantified spectral parameters (peak frequency and skewness index) that required a sufficiently strong gamma power induction in the spectrum for reliable estimation. Previous studies (Buzsáki and Wang, 2012; Fries et al., 2007; Tiesinga and Sejnowski, 2009; Vinck et al., 2013) have shown that neurons, and in particular pyramidal cells, display only weak locking to network population oscillations and further, that spike rates are often lower than the population gamma frequency. That is, single units regularly skip cycles. Consistently with this, the majority of units that we recorded exhibited only weakly oscillatory spike trains. The selected neurons (n = 5 in monkey S and n = 2 in monkey K) however, fired spikes on many gamma cycles and thus exhibited a strong periodic component in their spike trains. We must emphasize that these neurons are not representative of the whole population of cortical cells but rather of a *subclass* of cells that exhibit strong gamma periodicity. Here we show that these single units exhibited the same frequency modulations and shifts in spectral asymmetry as a function of contrast as those observed in population signals (LFP and MEG).

Gamma peak frequency as a robust parameter of the visual cortical network

The observation of contrast-dependent gamma frequency modulations fits with studies linking gamma frequency with other visual features such as stimulus size (Gieselmann and Thiele, 2008; Ray and

Maunsell, 2011), eccentricity (van Pelt and Fries, 2013), motion speed (Friedman-Hill et al., 2000; Swettenham et al., 2009), and motion direction (Feng et al., 2010). Gamma peak frequency, but not power, shows correlations with perceptual performance, cortical structure, and age, and has been shown to be highly heritable (Edden et al., 2009; Gaetz et al., 2012; Schwarzkopf et al., 2012; van Pelt et al., 2012). This emphasizes the importance of frequency measures in understanding human network function. Our results of strikingly similar contrast-induced modulation in gamma frequency in macaque LFP, single unit spike trains and human MEG (see regression slopes in Fig. 5) suggest that gamma frequency may be a spectral parameter that is robust across vastly different spatial scales (from single units to MEG) and thus provide a valuable tool in future investigations of human network function.

Power modulation of gamma oscillations by stimulus contrast

We observed luminance contrast-dependent gamma power modulation in the human MEG as well as in monkey LFP and single unit spike trains. In both human MEG and monkey LFP and spike trains, gamma power increased strongly in the lower- to-middle contrast ranges. However, we observed a substantial difference in the saturation of power at high contrasts. Whereas gamma power in the LFP and single-unit spike trains saturated (Figs. 3, 4), despite increasing spike rates (Fig. 3), MEG gamma power increased approximately linearly with contrast (Figs. 2, 4). The monotonic increase in induced gamma power with contrast in human MEG is in line with previous human MEG (Hall et al., 2005; Perry et al., 2014) and EEG (Koch et al., 2009) studies. In addition, the finding of non-monotonic increase in gamma power in V1 LFP is in line with two other studies in awake macaques (Bosman et al., 2012 (Suppl. Fig. S5); Ray and Maunsell, 2010), although mixed results are reported by studies in anaesthetized macaques (Henrie and Shapley (2005) report lack of saturation, whereas saturation is reported by Jia et al. (2013)).

Computational modeling of gamma-generating networks indicates that contrast-dependent modulation of gamma power with a non-monotonic component might arise by changing afferent E-drive to the excitatory–inhibitory spiking network (see Fig. 7B in Roberts et al., 2013), but the exact mechanisms underlying the relationship between power and contrast (or even excitatory input strength) are still not well established (Jadi and Sejnowski, 2014; Jia et al., 2013; Roberts et al., 2013). A possibly critical component for explaining non-monotonic power modulations is neural resonance (Moca et al., 2014), where there is enhancement of power at particular frequencies and not others. Optogenetic studies have provided experimental evidence for resonance in the gamma range (Cardin et al., 2009), with particular involvement of fast-spiking inhibitory neurons.

It is not clear what underlies the difference of the contrast — gamma power response function between MEG and macaque electrophysiological recordings. The observed difference in contrast response functions of power may be rooted in cross-species differences, in the different signals/recording techniques used or in the differences in experimental setups despite our efforts to minimize such differences. It could be the case that the differences observed are due to small differences in size and eccentricity of the stimuli. The most likely hypothesis in this vein is that a larger contribution of the transient pathway in the monkey, due to slightly more peripheral stimulus presentation than in the human participants, has led to greater saturation of gamma power with increased contrast in the monkeys. However, the typical saturation effect in the spiking rate seen for eccentric stimuli (Sclar et al., 1990) is not present in our data, which suggests that a much greater contribution of the transient system in the monkeys than in the humans is not a plausible explanation for the differences in contrast-induced power change seen between monkeys and humans. Hence, cross-species differences cannot be ruled out, although such conclusion would require an exclusion of the possibility that differences in the contrast-dependency of gamma power between monkeys and humans are due to the different manners in which LFP and MEG signals are generated. The LFP is comparatively local while the MEG signal is based on aggregation over space, causing it to be dominated by a much larger relative contribution of synchronous compared to asynchronous meso-sources (Nunez and Srinivasan, 2006). For instance, it is possible that increasing contrast may result in increasing longer range spatial synchronization, which may outweigh local saturation effects as picked up with LFP. The relevance of aggregation effects is supported by a recent electrophysiological study in monkeys from Musall et al. (2014) in which spatial LFP synchrony across electrodes was shown to be a major determinant of simultaneously recorded EEG power. If this can be confirmed, it would indicate that LFP power and MEG power are complementary signals.

A comparison of the effect of contrast modulations on spike rates, LFP frequency and LFP power

Luminance contrast, a central visual parameter, is a proxy for excitatory drive (E-drive) to the primary visual cortex, increasing the spike rates of the majority of neurons in the retino-geniculate pathway as well as early visual cortex (Contreras and Palmer, 2003; Sclar et al., 1990). Plausible biophysical mechanisms for E-drive dependent oscillation frequency shift have been described in a number of computational modeling studies of gamma generating networks (Brunel and Wang, 2003; Roberts et al., 2013; Traub et al., 1996).

The theoretical models predict a close link between the precise gamma frequency and the overall excitation state of the underlying network. This view is supported by our finding of similar monotonic increases in gamma frequency and multi-unit spike rates as a function of stimulus contrast. Similar observations have been reported by other macaque V1 studies (Jia et al., 2013; Ray and Maunsell, 2010). In contrast to LFP *gamma frequency*, LFP *gamma power* saturated or even decreased at high contrasts despite monotonically increasing spike rates. This indicates that LFP gamma power was not tightly related to spiking in the majority of recorded neurons (Logothetis et al., 2001). The LFP predominantly reflects synaptic, rather than spiking events, particularly when these result in synchronous fluctuations of the membrane potentials across many neurons (Buzsáki et al., 2012). Thus, changes in synchronization between neurons can change the power of the LFP in the absence of changes in firing rate. Thus, one possibility for the observed dissociation is that while firing rates continue to go up with increased contrast (increased afferent excitatory drive), locally neurons become less synchronous resulting in lower LFP power.

Contrast-induced modulations of the asymmetry of gamma spectral distributions

We have observed in both human and macaque visual cortex and at all measurement scales (MEG, LFP and single-unit spikes) systematic asymmetries in the gamma spectral distribution (right/left skewness of the gamma power spectra), which were modulated by stimulus contrast. To our knowledge, this has not been reported so far. The shape of the gamma power distribution may reveal further properties of the underlying oscillatory process. Asymmetry may be a meaningful characteristic of the dynamical process underlying gamma; an idea that is supported by its systematic changes with stimulus contrast.

The asymmetry measure we used here was derived from baseline-normalized spectra. One concern is that because the baseline spectra are characterized by a 1/f power scaling (Miller et al., 2009; Pritchard, 1992), the baseline-normalization could lead to asymmetry of the gamma power distribution artificially. Several observations contradict this possibility. First, the asymmetries observed in spike train data are similar to the ones observed in LFP/MEG data, yet, spike train power spectra in the baseline were not characterized by 1/f characteristics. Second, similar effect ranges between MEG and LFP were observed, although the power ratio of induced power to baseline power was considerably different (gamma peak power in human MEG ~ 1.5–2

and monkey LFP ~ 5–10). We would expect that a lower power ratio would yield larger asymmetries; however, this was not observed. Further, the 1/f characteristic of the spectra in the baseline cannot explain the reversal of asymmetry at higher contrasts.

As to the underlying causes of the empirically-observed spectral asymmetries, these are still unknown. The moment-to-moment variation in gamma oscillation power and frequency has been reported as being highly complex (Burns et al., 2010, 2011) and shown to be modulated by multiple other rhythmic neuronal processes (Bosman et al., 2009; Jensen and Colgin, 2007; Lakatos et al., 2005; Osipova et al., 2008; Schroeder and Lakatos, 2009). Slow microsaccadic rhythms that modulate the gamma oscillations may also contribute to such asymmetries. A further possible factor for explaining power asymmetries may be neural resonance (Moca et al., 2014), where particular frequencies lead to enhancement of power. The understanding of the dynamical properties of gamma oscillations in vivo, their appropriate quantification as well as their neurocomputational significance are still far from being fully elucidated (Burns et al., 2010, 2011; Cohen, 2014; Lakatos et al., 2005; Roberts et al., 2013; Schroeder and Lakatos, 2009). Future detailed modeling work will need to investigate possible mechanisms behind gamma power asymmetries.

Measurement sensitivity of gamma oscillations with non-invasive MEG in humans

A prominent difference between human MEG measured gamma power and gamma power in macaque V1 measured microelectrodes is the large difference in overall induced power indicating low measurement sensitivity (Hadjipapas et al., 2007; Muthukumaraswamy et al., 2010; Swettenham et al., 2009). This was true despite the fact that stimulus parameters were optimized for the MEG experiment. The low gamma SNR becomes exacerbated for small cortical stimulus representations, i.e., when eccentricity is high (Swettenham et al., 2009; van Pelt and Fries, 2013) or stimulus size is small (Gieselmann and Thiele, 2008; Perry et al., 2013). Hence, to increase SNR in the MEG experiment the stimuli were placed at a smaller eccentricity and presented in a square aperture to restrict their presentation to one hemifield. Additionally, the contrast manipulation was shifted to higher contrast values. Furthermore, we used a beamforming technique to improve the localization and SNR of gamma oscillations.

Still, we did not observe significant MEG gamma power at the lowest contrast used (20% contrast), yet we observed substantial induced power in the gamma range as low as 6% contrast in both monkeys. Reports in rhesus macaque studies have indicated that gamma oscillations can even be detected in conditions of no visual stimulation (Spaak et al., 2012; Vinck et al., 2013).

A critical point to consider is the different scale of measurement involved in the MEG compared to microelectrodes. Even when the MEG signal stems from a well-reconstructed source, it still reflects aggregated neural activity of orders of magnitude larger than the local field potential and spiking activity measured by microelectrodes. The MEG signal is therefore very sensitive to the neural synchrony across larger cortical distances. However, the synchrony of gamma oscillation has been shown to be spatially limited (Eckhorn et al., 2004). Hence, the more local spatial scale of gamma-band synchronization makes MEG/EEG recordings less sensitive to gamma oscillations.

Another important point to consider is the cellular/cortical origin of the MEG signal. Due to their dominant somatodendritic axis, pyramidal neurons collectively give rise to current summation and resulting fields that can be measured at a distance from their source (so-called 'open-field' configurations; Buzsáki et al., 2012; Murakami and Okada, 2006). The MEG signal is thought to predominantly reflect currents running along the somatodendritic axis that arise from opposing charges at the apical dendrites and the soma of pyramidal neurons (Hämäläinen et al., 1993). Quantitative studies taking into account neuronal and dendritic geometry indicate that layer V neurons make a larger contribution than layer II–III neurons (Murakami and Okada, 2006). This is likely because superficial layer pyramidal neurons are smaller and have much shorter dendrites, a smaller separation between soma and dendrites and therefore a less pronounced open field configuration, and a lesser dipole moment (Jones et al., 2007). Several macaque studies have indicated that the strongest source of gamma oscillations are observed in the superficial and middle cortical layers (Buffalo et al., 2011; Roberts et al., 2013; Xing et al., 2012). Hence, assuming a layer dependency in the human that is similar to that in the macaque, the contribution of superficial layer gamma to the MEG signal would be small compared to that from the layer V neurons (Jones et al., 2007).

Conclusions

Our results in primary visual cortex show that contrast-induced modulations in gamma frequency and gamma spectral asymmetry in human MEG matched the results from macaque microelectrode recordings, including results from well-isolated single unit recordings. However, there were discrepancies in gamma power. The latter is more complex to understand not least because gamma power in the MEG depends on longer-range spatial synchrony. Thus, while oscillation frequency was found to be more stable across spatial scales (Fig. 5) and to convey similar information about the stimulus, LFP and MEG power seemed to convey complementary information. Our results and theoretical considerations on spatial aggregation lead us to the conclusion that in spite of many similar findings between human MEG and monkey LFP/spike data, there are also differences and one needs to be prudent in translating macaque microelectrode findings to human MEG, in particular for oscillatory power. To fully understand the similarities and differences in findings from invasive recordings in monkeys and non-invasive measurements in humans, and in the underlying cortical mechanisms, future work will need to involve detailed modeling studies taking into account contributions of functional and structural properties of single neurons, of different layers, of within- and across-layer connectivity, as well as effects of overall network size and stimulus drive.

Acknowledgments

Research was supported by NWO VICI grant to P.D.W. (#453-04-002), NWO VENI grant to M.R. (#451-09-025), European Union Seventh Framework Programme (FP7/2007-2013) under grant agreement no. 604102 (Human Brain Project) to P.D.W. and A.H., Maastricht University Graduate funding in cooperation with the Donders Institute for Brain Behavior and Cognition to P.D.W. All data collection and initial analysis was conducted at the Donders Institute for Brain, Cognition and Behaviour.

Conflict of interest

The authors declare no competing financial interests.

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